Scaling invariance of the angular lower-tail coefficient and an application to the Pareto-mixture copula

Definition. Let C be a copula on $[0,1]^2$. For $\delta \in (0,2)$ and $\gamma > 0$, define

$$\kappa_L(\delta, \gamma; C) = \lim_{u \downarrow 0} \frac{\log C\big(\gamma u^{\delta}, \gamma^{-1} u^{2-\delta}\big)}{\log u},$$

whenever the limit exists.

The purpose of this note is twofold:

1. to prove that, under a minimal existence assumption, $\kappa_L(\delta, \gamma; C)$ does not depend on the scaling parameter γ;
2. to verify that the Pareto-mixture copula C_{PM} satisfies this property.

1. A general scaling-invariance theorem

For the general statement below, no smoothness assumption on C is needed beyond the defining properties of a copula. The only regularity condition is existence of the limit for one reference scaling.

Theorem 1. *Fix $\delta \in (0,2)$ and let C be a copula. Assume that there exists some $\tilde{\gamma} > 0$ such that*

$$k := \lim_{u \downarrow 0} \frac{\log C\big(\tilde{\gamma} u^{\delta}, \tilde{\gamma}^{-1} u^{2-\delta}\big)}{\log u}$$

exists as a finite real number. Then, for every $\gamma > 0$,

$$\lim_{u \downarrow 0} \frac{\log C\big(\gamma u^{\delta}, \gamma^{-1} u^{2-\delta}\big)}{\log u} = k.$$

Hence $\kappa_L(\delta, \gamma; C)$ is independent of γ.

Proof. Fix $\delta \in (0,2)$ and $\gamma > 0$. Let

$$r := \frac{\gamma}{\tilde{\gamma}} > 0, \qquad c_- := \min\Big\{r^{1/\delta},\, r^{-1/(2-\delta)}\Big\}, \qquad c_+ := \max\Big\{r^{1/\delta},\, r^{-1/(2-\delta)}\Big\}.$$

Then $c_- > 0$ and $c_+ > 0$.

For every $u > 0$,

$$\frac{\tilde{\gamma}(c_- u)^\delta}{\gamma u^\delta} = \frac{\tilde{\gamma} c_-^\delta}{\gamma} = \frac{c_-^\delta}{r} \le 1,$$

because $c_- \le r^{1/\delta}$. Likewise,

$$\frac{\tilde{\gamma}^{-1}(c_- u)^{2-\delta}}{\gamma^{-1} u^{2-\delta}} = \frac{\gamma}{\tilde{\gamma}} c_-^{2-\delta} = r\, c_-^{2-\delta} \le 1,$$

because $c_- \le r^{-1/(2-\delta)}$. Hence

$$\tilde{\gamma}(c_- u)^\delta \le \gamma u^\delta, \qquad \tilde{\gamma}^{-1}(c_- u)^{2-\delta} \le \gamma^{-1} u^{2-\delta}. \tag{1}$$

Similarly, since $c_+ \ge r^{1/\delta}$ and $c_+ \ge r^{-1/(2-\delta)}$,

$$\tilde{\gamma}(c_+ u)^\delta \ge \gamma u^\delta, \qquad \tilde{\gamma}^{-1}(c_+ u)^{2-\delta} \ge \gamma^{-1} u^{2-\delta}. \tag{2}$$

A copula is coordinatewise nondecreasing. Therefore (1) and (2) imply that, for every $u > 0$,

$$C\big(\tilde{\gamma}(c_- u)^\delta, \tilde{\gamma}^{-1}(c_- u)^{2-\delta}\big) \le C\big(\gamma u^\delta, \gamma^{-1} u^{2-\delta}\big) \le C\big(\tilde{\gamma}(c_+ u)^\delta, \tilde{\gamma}^{-1}(c_+ u)^{2-\delta}\big). \tag{3}$$

Because the reference limit is assumed to exist as a finite real number, there exists $v_0 > 0$ such that

$$C\big(\tilde{\gamma} v^\delta, \tilde{\gamma}^{-1} v^{2-\delta}\big) > 0 \qquad \text{for all } 0 < v < v_0.$$

Choose $u_0 > 0$ so small that $c_+ u < v_0$ whenever $0 < u < u_0$. Then all three terms in (3) are strictly positive for $u \in (0, u_0)$. Since log is increasing on $(0,\infty)$,

$$\log C\big(\tilde{\gamma}(c_- u)^\delta, \tilde{\gamma}^{-1}(c_- u)^{2-\delta}\big) \le \log C\big(\gamma u^\delta, \gamma^{-1} u^{2-\delta}\big) \le \log C\big(\tilde{\gamma}(c_+ u)^\delta, \tilde{\gamma}^{-1}(c_+ u)^{2-\delta}\big).$$

For u sufficiently small, $\log u < 0$. Dividing by $\log u$ reverses the inequalities:

$$\frac{\log C\big(\tilde{\gamma}(c_- u)^\delta, \tilde{\gamma}^{-1}(c_- u)^{2-\delta}\big)}{\log u} \ge \frac{\log C\big(\gamma u^\delta, \gamma^{-1} u^{2-\delta}\big)}{\log u} \ge \frac{\log C\big(\tilde{\gamma}(c_+ u)^\delta, \tilde{\gamma}^{-1}(c_+ u)^{2-\delta}\big)}{\log u}. \tag{4}$$

Now fix any $c > 0$ and define

$$A_c(u) := \frac{\log C\big(\tilde{\gamma}(cu)^{\delta}, \tilde{\gamma}^{-1}(cu)^{2-\delta}\big)}{\log u}.$$

For small $u > 0$,

$$A_c(u) = \frac{\log C\big(\tilde{\gamma}(cu)^{\delta}, \tilde{\gamma}^{-1}(cu)^{2-\delta}\big)}{\log(cu)} \cdot \frac{\log(cu)}{\log u}.$$

As $u \downarrow 0$, we have $cu \downarrow 0$, so by the definition of k,

$$\frac{\log C\big(\tilde{\gamma}(cu)^{\delta}, \tilde{\gamma}^{-1}(cu)^{2-\delta}\big)}{\log(cu)} \longrightarrow k.$$

Also,

$$\frac{\log(cu)}{\log u} = 1 + \frac{\log c}{\log u} \longrightarrow 1 \qquad (u \downarrow 0),$$

because $\log u \to -\infty$. Therefore

$$A_c(u) \longrightarrow k \qquad (u \downarrow 0). \tag{5}$$

Applying (5) with $c = c_-$ and $c = c_+$ shows that both outer terms in (4) converge to k. Hence, by the squeeze theorem,

$$\lim_{u \downarrow 0} \frac{\log C\big(\gamma u^{\delta}, \gamma^{-1} u^{2-\delta}\big)}{\log u} = k.$$

This proves the claim. □

Remark 2. *Theorem 1 is the precise formulation of the desired scaling invariance. For each fixed $\delta \in (0, 2)$, it is enough to know that the limit exists for one value of γ; then it automatically exists for every $\gamma > 0$ and has the same value.*

2. The Pareto-mixture copula

We now verify the assumption of Theorem 1 for the proposed Pareto-mixture copula.

Model. Assume throughout that $\mathrm{Pa}(s)$ denotes the standard Pareto law on $[1,\infty)$ with survival function

$$\mathbb{P}(Z > z) = z^{-s}, \qquad z \geq 1.$$

Let

$$X_1 = \frac{R}{R^*}\frac{H_1}{H_1^*}, \qquad X_2 = \frac{R}{R^*}\frac{H_2}{H_2^*},$$

where

$$R \sim \mathrm{Pa}(\beta), \quad R^* \sim \mathrm{Pa}(\alpha), \quad H_1 \sim \mathrm{Pa}(b_1), \quad H_1^* \sim \mathrm{Pa}(a_1), \quad H_2 \sim \mathrm{Pa}(b_2), \quad H_2^* \sim \mathrm{Pa}(a_2),$$

all mutually independent, and all shape parameters are strictly positive.

Since (X_1, X_2) has continuous margins, its copula is well defined; we denote it by C_{PM}.

2.1. Logarithmic representation

Set

$$A := \log R, \quad A^* := \log R^*, \quad B_i := \log H_i, \quad B_i^* := \log H_i^* \qquad (i = 1, 2).$$

Then

$$A \sim \mathrm{Exp}(\beta), \qquad A^* \sim \mathrm{Exp}(\alpha), \qquad B_i \sim \mathrm{Exp}(b_i), \qquad B_i^* \sim \mathrm{Exp}(a_i),$$

independently, and

$$-\log X_i = (A^* - A) + (B_i^* - B_i).$$

Define

$$U_0 := A^* - A, \qquad U_i := B_i^* - B_i, \qquad i = 1, 2.$$

Then

$$-\log X_i = U_0 + U_i, \qquad i = 1, 2.$$

A direct convolution gives, for $s \in \mathbb{R}$,

$$f_{U_0}(s) = \frac{\alpha\beta}{\alpha + \beta} \begin{cases} e^{-\alpha s}, & s \geq 0, \\ e^{\beta s}, & s < 0, \end{cases}$$

and for each $i = 1, 2$,

$$\mathbb{P}(U_i > t) = \frac{b_i}{a_i + b_i} e^{-a_i t}, \qquad t \geq 0.$$

Consequently there exist constants $0 < k_i \leq K_i < \infty$ such that, for all $z \in \mathbb{R}$,

$$k_i e^{-a_i z_+} \leq \mathbb{P}(U_i \geq z) \leq K_i e^{-a_i z_+}, \qquad z_+ := \max\{z, 0\}. \tag{6}$$

2.2. Marginal lower-tail exponents

Let F_{X_i} denote the distribution function of X_i, and define

$$m_i := \min\{\alpha, a_i\}, \qquad i = 1, 2.$$

Lemma 3. *For each $i = 1, 2$,*

$$-\log F_{X_i}(e^{-t}) = m_i\, t + o(t) \qquad (t \to \infty).$$

Equivalently,

$$F_{X_i}(e^{-t}) = \exp\bigl(-m_i t + o(t)\bigr).$$

Proof. Fix $i \in \{1, 2\}$. Since $X_i \le e^{-t}$ if and only if $U_0 + U_i \ge t$,

$$F_{X_i}(e^{-t}) = \mathbb{P}(U_0 + U_i \ge t) = \int_{\mathbb{R}} \mathbb{P}(U_i \ge t - s)\, f_{U_0}(s)\, ds.$$

Split the integral into three parts:

$$(-\infty, 0], \qquad [0, t], \qquad [t, \infty).$$

Using (6) and the formula for f_{U_0}, we obtain

$$F_{X_i}(e^{-t}) \le C_1 e^{-a_i t} + C_2 \int_0^t e^{-\alpha s} e^{-a_i(t-s)}\, ds + C_3 e^{-\alpha t},$$

for suitable positive constants C_1, C_2, C_3 independent of t.

Hence

$$F_{X_i}(e^{-t}) \le C \begin{cases} e^{-a_i t}, & a_i < \alpha, \\ (1+t)e^{-\alpha t}, & a_i = \alpha, \\ e^{-\alpha t}, & \alpha < a_i, \end{cases}$$

for some constant $C > 0$.

For the lower bound, we consider three cases.

Case 1: $a_i < \alpha$. Integrating over $s \in [0, 1]$ gives

$$F_{X_i}(e^{-t}) \ge c \int_0^1 e^{-\alpha s} e^{-a_i(t-s)}\, ds \ge c' e^{-a_i t}.$$

Case 2: $\alpha < a_i$. Integrating over $s \in [t, t+1]$ gives

$$F_{X_i}(e^{-t}) \ge c \int_t^{t+1} e^{-\alpha s}\, ds \ge c' e^{-\alpha t}.$$

Case 3: $\alpha = a_i$. Integrating over $s \in [0,t]$ gives

$$F_{X_i}(e^{-t}) \geq c \int_0^t e^{-\alpha s} e^{-\alpha(t-s)}\, ds = c\, t\, e^{-\alpha t}.$$

Combining the upper and lower bounds yields

$$F_{X_i}(e^{-t}) = \exp\bigl(-m_i t + o(t)\bigr),$$

which is the desired conclusion. □

Lemma 4. *Let F be continuous and strictly increasing near 0, and suppose that for some $m > 0$,*

$$-\log F(e^{-t}) = mt + o(t) \qquad (t \to \infty).$$

Then

$$-\log F^{-1}(u) = \frac{1}{m} \log\frac{1}{u} + o\left(\log\frac{1}{u}\right) \qquad (u \downarrow 0).$$

Proof. Let $y(u) := -\log F^{-1}(u)$. Since $F(F^{-1}(u)) = u$,

$$-\log u = -\log F(e^{-y(u)}) = m\, y(u) + o\bigl(y(u)\bigr).$$

As $u \downarrow 0$, we have $F^{-1}(u) \downarrow 0$, hence $y(u) \to \infty$. Therefore

$$y(u) = \frac{1}{m} \log\frac{1}{u} + o\left(\log\frac{1}{u}\right).$$

□

Applying Lemmas 3 and 4 to F_{X_1} and F_{X_2}, we obtain:

$$-\log F_{X_i}^{-1}(u) = \frac{1}{m_i} \log\frac{1}{u} + o\left(\log\frac{1}{u}\right), \qquad i = 1, 2. \tag{7}$$

2.3. Existence of the angular lower-tail limit for C_{PM}

Define

$$q_1 := \frac{\delta}{m_1}, \qquad q_2 := \frac{2-\delta}{m_2}.$$

Proposition 5. *For every $\delta \in (0,2)$, the limit*

$$\kappa_L(\delta, 1; C_{\mathrm{PM}}) = \lim_{u \downarrow 0} \frac{\log C_{\mathrm{PM}}(u^\delta, u^{2-\delta})}{\log u}$$

exists as a finite real number.

More precisely, let

$$J(q_1, q_2) := \inf_{s \geq 0} \Big\{ \alpha s + a_1 (q_1 - s)_+ + a_2 (q_2 - s)_+ \Big\}.$$

Then

$$\kappa_L(\delta, 1; C_{\mathrm{PM}}) = J(q_1, q_2).$$

Proof. Let

$$u = e^{-t}, \qquad t \to \infty.$$

Define

$$y_1(t) := -\log F_{X_1}^{-1}(u^\delta), \qquad y_2(t) := -\log F_{X_2}^{-1}(u^{2-\delta}).$$

By (7),

$$y_1(t) = q_1 t + o(t), \qquad y_2(t) = q_2 t + o(t). \tag{8}$$

Now

$$C_{\mathrm{PM}}(u^\delta, u^{2-\delta}) = \mathbb{P}\big(X_1 \leq F_{X_1}^{-1}(u^\delta),\, X_2 \leq F_{X_2}^{-1}(u^{2-\delta})\big).$$

Since $X_i \leq e^{-y_i(t)}$ is equivalent to $U_0 + U_i \geq y_i(t)$, we obtain

$$C_{\mathrm{PM}}(u^\delta, u^{2-\delta}) = \mathbb{P}\big(U_0 + U_1 \geq y_1(t),\, U_0 + U_2 \geq y_2(t)\big).$$

Conditioning on U_0 gives

$$C_{\mathrm{PM}}(u^\delta, u^{2-\delta}) = \int_{\mathbb{R}} \mathbb{P}\big(U_1 \geq y_1(t) - s\big)\, \mathbb{P}\big(U_2 \geq y_2(t) - s\big)\, f_{U_0}(s)\, ds. \tag{9}$$

We first treat the case $q_1 < q_2$. Then by (8), for all sufficiently large t,

$$y_1(t) \leq y_2(t).$$

Split the integral in (9) into four intervals:

$$(-\infty, 0], \qquad [0, y_1(t)], \qquad [y_1(t), y_2(t)], \qquad [y_2(t), \infty).$$

Using (6) and the explicit density of U_0, we obtain

$$C_{\mathrm{PM}}(u^\delta, u^{2-\delta}) \leq C\Big(e^{-A_t} + e^{-B_t} + e^{-D_t}\Big),$$

where

$$A_t := a_1 y_1(t) + a_2 y_2(t), \qquad B_t := \alpha y_1(t) + a_2\big(y_2(t) - y_1(t)\big), \qquad D_t := \alpha y_2(t),$$

and $C > 0$ is independent of t.

Indeed:

$$\begin{aligned}
&\int_{-\infty}^{0} \cdots \, ds \le C e^{-a_1 y_1(t) - a_2 y_2(t)}, \\
&\int_{0}^{y_1(t)} \cdots \, ds \le C\Big(e^{-a_1 y_1(t) - a_2 y_2(t)} + e^{-\alpha y_1(t) - a_2(y_2(t) - y_1(t))}\Big), \\
&\int_{y_1(t)}^{y_2(t)} \cdots \, ds \le C\Big(e^{-\alpha y_1(t) - a_2(y_2(t) - y_1(t))} + e^{-\alpha y_2(t)}\Big), \\
&\int_{y_2(t)}^{\infty} \cdots \, ds \le C e^{-\alpha y_2(t)}.
\end{aligned}$$

For the lower bound, we integrate over the unit intervals $[0, 1]$, $[y_1(t), y_1(t)+1]$, and $[y_2(t), y_2(t)+1]$. Since the density f_{U_0} is strictly positive on compact sets and (6) gives strictly positive lower bounds for the relevant tail probabilities, there exists $c > 0$ such that

$$C_{\mathrm{PM}}(u^\delta, u^{2-\delta}) \ge c \max\Big\{e^{-A_t},\, e^{-B_t},\, e^{-D_t}\Big\}.$$

Combining upper and lower bounds,

$$-\log C_{\mathrm{PM}}(u^\delta, u^{2-\delta}) = \min\{A_t, B_t, D_t\} + O(1).$$

Dividing by t and using (8),

$$-\frac{1}{t}\log C_{\mathrm{PM}}(u^\delta, u^{2-\delta}) \longrightarrow \min\Big\{a_1 q_1 + a_2 q_2,\, \alpha q_1 + a_2(q_2 - q_1),\, \alpha q_2\Big\}.$$

A direct check shows that this minimum equals

$$\inf_{s \ge 0}\Big\{\alpha s + a_1(q_1 - s)_+ + a_2(q_2 - s)_+\Big\} = J(q_1, q_2).$$

If $q_2 < q_1$, the same argument with the roles of indices 1 and 2 exchanged yields the same conclusion.

Finally, if $q_1 = q_2$, then (8) implies

$$y_1(t) - y_2(t) = o(t),$$

so the two possible orderings differ only by $o(t)$ in the exponent. Hence the same limit still exists and equals $J(q_1, q_2)$.

Since $\log u = -t$, we conclude that

$$\frac{\log C_{\mathrm{PM}}(u^\delta, u^{2-\delta})}{\log u} = -\frac{1}{t} \log C_{\mathrm{PM}}(u^\delta, u^{2-\delta}) \longrightarrow J(q_1, q_2).$$

Therefore

$$\kappa_L(\delta, 1; C_{\mathrm{PM}}) = J(q_1, q_2),$$

and the limit exists as a finite real number. □

2.4. Independence of the scaling parameter

We can now combine Proposition 5 with Theorem 1.

Theorem 6. *For the Pareto-mixture copula C_{PM} and every $\delta \in (0, 2)$,*

$$\kappa_L(\delta, \gamma; C_{\mathrm{PM}})$$

exists for every $\gamma > 0$ and is independent of γ.

In fact, if

$$m_1 = \min\{\alpha, a_1\}, \qquad m_2 = \min\{\alpha, a_2\}, \qquad q_1 = \frac{\delta}{m_1}, \qquad q_2 = \frac{2-\delta}{m_2},$$

then for every $\gamma > 0$,

$$\kappa_L(\delta, \gamma; C_{\mathrm{PM}}) = \inf_{s \geq 0} \Big\{ \alpha s + a_1 (q_1 - s)_+ + a_2 (q_2 - s)_+ \Big\}.$$

Proof. By Proposition 5, for each fixed $\delta \in (0, 2)$ the limit defining $\kappa_L(\delta, 1; C_{\mathrm{PM}})$ exists as a finite real number. Therefore the hypothesis of Theorem 1 is satisfied with $\tilde{\gamma} = 1$. Applying Theorem 1 yields

$$\kappa_L(\delta, \gamma; C_{\mathrm{PM}}) = \kappa_L(\delta, 1; C_{\mathrm{PM}}) \qquad \text{for all } \gamma > 0.$$

The explicit formula then follows from Proposition 5. □

Conclusion

Under the minimal regularity condition that, for each fixed $\delta \in (0, 2)$, the logarithmic limit exists for one reference scaling $\tilde{\gamma} > 0$, the angular lower-tail coefficient

$$\kappa_L(\delta, \gamma; C) = \lim_{u \downarrow 0} \frac{\log C\big(\gamma u^\delta, \gamma^{-1} u^{2-\delta}\big)}{\log u}$$

is independent of γ.

For the Pareto-mixture copula C_{PM}, this condition is satisfied for every $\delta \in (0, 2)$; hence $\kappa_L(\delta, \gamma; C_{\mathrm{PM}})$ exists for all $\gamma > 0$ and is indeed scaling-invariant.